

Answers

1. Lerp between 2 direction vectors always results in unit length vectors.

Answer: False

2. Lerp is the most performant vector interpolation method.

Answer: True

3. Nlerp can potentially result in a zero vector.

Answer: True

4. Nlerp is also known as normalized lerp.

Answer: True

5. Using nlerp is often the ideal choice for interpolating vectors.

Answer: True

6. After applying nlerp, the resulting vector will always be of unit length.

Answer: True

7. Slerp stands for simple linear interpolation.

Answer: False

8. Slerp travels over the surface of the sphere, providing constant angular velocity.

Answer: True

9. Slerp is more performant than nlerp.

Answer: False

10. If 2 vectors are facing away from each other at 180 degrees, which of the 3 lerp methods (lerp, nlerp, slerp) can result in zero vectors?

Answer: Both lerp and nlerp

11. How does nlerp differ from a simple lerp?

Answer: Nlerp is a normalized lerp, where after performing the lerp, the resulting vector is normalized to make it unit length.

12. In the context of 3D graphics, why are interpolated normals often renormalized?

Answer: *To ensure they remain unit vectors, preserving their directionality without introducing artifacts.*

13. Why is nlerp often used, despite its drawbacks?

Answer: *Nlerp is more computationally efficient than slerp and in most cases, it's just "good enough". Especially in games where it's often the case that exact precision isn't critical.*

14. Why is slerp preferred when exact precision in interpolation is required?

Answer: *Slerp maintains constant angular velocity and accounts for the spherical nature of rotations, making transitions appear uniform.*